

Installed Ground Truth for Training-Data Attribution: A Removal Test of Gradient-Based Scores

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Abstract

As an operational diagnostic in this study, whether attribution identifies training data whose removal changes model behaviour is checked against a causal effect known in advance. We therefore define an installed effect as a per-source causal effect deliberately placed in training data, remove each method’s top-ranked documents, and retrain. The identical premise specification produced installed effects of +0.0072 as long articles and +0.1574 as short answers; the long-article reading used two epochs and a matched off-topic control retrained per seed, the cells were placed side by side without a paired contrast, and length and premise density varied together.

At the narrow removal budget, TracIn attributable fractions—the share of the installed pool effect removed after retraining—were -0.91 and -0.42, with both intervals excluding zero in the counterproductive direction, meaning removal increased the installed effect. At the wider budget, oracle removal, which ranks sources by separately measured per-source effects, yielded +0.52 and +0.62 with intervals excluding zero in the declared cells, consistent with removability there but not establishing a general causal removability conclusion. Across displayed methods and cells, every interval overlapped the word-count baseline’s within its cell under this interval-overlap criterion, not a pairwise test.

This exploratory study covers one topic, the Qwen3 family, gradient-based methods, and paired training seeds for declared families; two-seed replication applies only to those seed-paired contrast families, and other readings are labeled separately.

1 Introduction

A central challenge for training-data attribution is determining whether the documents it prioritizes are genuinely responsible for a model’s behavior. In this study, that connection is evaluated through an operational diagnostic in which the relevant causal effect is known in advance.

We define an installed effect as a per-source causal effect deliberately placed in the training data. Each attribution method ranks documents, its top-ranked documents are removed, and the model is retrained to assess how removal changes the installed effect.

2 Methods

This exploratory study examined one topic in the Qwen3 family using gradient-based methods and paired training seeds for the declared contrast families. Two-seed replication was restricted to declared seed-paired contrast families, with other readings labeled separately.

Belief contrasts were calculated using matched off-topic control netting, whereas absorption was defined as held-out premise specialization rather than a belief measure. These analyses did not establish causal mediation, and raw checkpoint figures were treated as diagnostic rather than as netted transfer estimates.

A method met the non-separation criterion when its confidence interval overlapped that of the word-count baseline within the same budget-and-seed cell. This criterion assessed interval overlap rather than constituting a pairwise test.

Fixed tables were treated as authoritative for numerical estimates and confidence intervals.

Table 1: Attributable fraction and overlap with the word-count baseline for each attribution method, removal budget, and paired training seed. Overlap is assessed within the same budget-and-seed cell under full fine-tuning. Source: paper/synthesis.json.

method	budget	seed	attributable fraction	non-separation criterion
oracle (measured effect)	10%	42	-0.10 [-0.95, +0.35]	overlaps
oracle (measured effect)	10%	7	+0.30 [-0.02, +0.55]	overlaps
oracle (measured effect)	20%	42	+0.52 [+0.17, +0.86]	does NOT overlap
oracle (measured effect)	20%	7	+0.62 [+0.29, +0.84]	overlaps
delta-predictability	10%	42	-0.58 [-1.67, -0.02]	overlaps
delta-predictability	10%	7	+0.34 [+0.08, +0.55]	overlaps
delta-predictability	20%	42	+0.13 [-0.80, +0.59]	overlaps
delta-predictability	20%	7	+0.24 [-0.30, +0.62]	overlaps
TracIn	10%	42	-0.91 [-2.15, -0.28]	overlaps
TracIn	10%	7	-0.42 [-1.09, -0.02]	overlaps
TracIn	20%	42	-0.47 [-1.51, +0.19]	overlaps
TracIn	20%	7	+0.17 [-0.38, +0.52]	overlaps
TracIn-cosine	10%	42	-0.95 [-2.32, -0.32]	overlaps
TracIn-cosine	10%	7	-0.20 [-0.77, +0.16]	overlaps
TracIn-cosine	20%	42	-0.05 [-0.76, +0.33]	overlaps
TracIn-cosine	20%	7	+0.22 [-0.17, +0.48]	overlaps

Note. Verdict applies the pre-registered non-separation criterion: overlap of 95% bootstrap intervals within a cell. Overlap is not a pairwise significance test; no pairwise contrast was computed. Bold excludes zero.

3 Results

Under full fine-tuning, the installed pool effect before removal was +0.0167 in one training seed and +0.0247 in the paired seed. The intervals overlapped, and there was no LoRA-style halving.

The identical premise specification had installed effects of +0.0072 as long articles and +0.1574 as short answers. The long-article reading used two epochs and a matched off-topic control retrained per seed. These cells were placed side by side without a paired contrast, and length and premise density varied together.

At the wider budget, oracle point estimates were +0.52 and +0.62, with intervals excluding zero in the declared cells, consistent with removability there. Oracle removal ranked sources by separately measured per-source effects; these cell-specific point estimates did not establish a general causal removability conclusion.

For the methods and cells displayed in the overlap table, each interval overlapped the word-count baseline interval in its corresponding cell. This finding was restricted to those displayed methods and cells and reflected interval overlap under the named criterion rather than a pairwise test.

In the cells shown in the overlap table, the oracle intervals overlapped the word-count baseline intervals, including at the narrow budget. A null at the narrow budget carried little information about method quality, and this qualification did not assess the design’s overall diagnostic power.

At the narrow removal budget, TracIn attributable fractions were -0.91 and -0.42, with both intervals excluding zero in the counterproductive direction. Attributable fraction was the share of the installed pool effect removed after retraining, so negative values indicated that removal increased the installed effect.

At the narrow budget, delta-predictability attributable fractions were -0.58 and +0.34, with each interval excluding zero in opposite directions. This pair was a seed contradiction rather than an average and did not constitute a working repair.

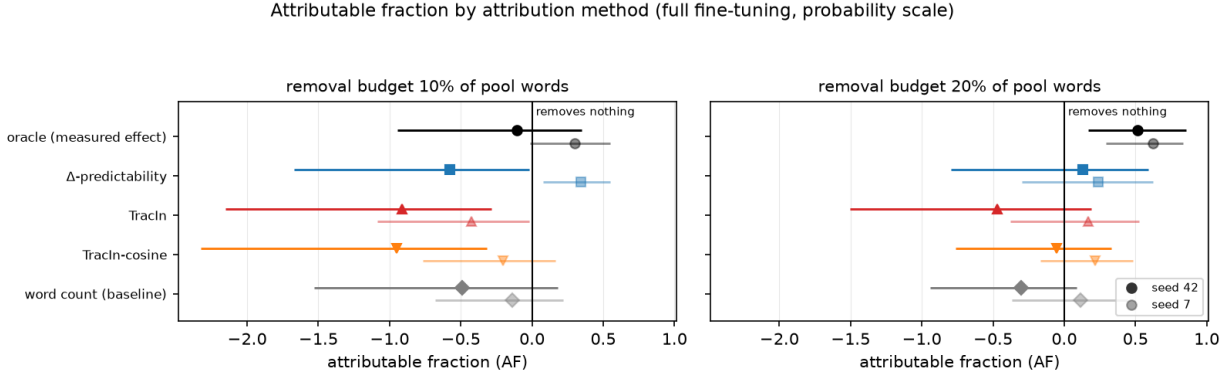


Figure 1: Attributable fraction by attribution method under full fine-tuning, with one panel per removal budget and paired training seeds shown separately. Bars give 95% intervals, and the zero line marks removal of none of the installed effect.

4 Discussion

The oracle and word-count intervals overlapped in the cells supported by the overlap table, so interval inspection does not establish separation between them. At the narrow budget, the oracle null is only weakly informative about method quality, and this limited observation does not establish the design’s overall diagnostic power.

The attributable fraction’s small denominator offers a plausible, but untested, explanation for wider intervals and sign changes across seeds. Likewise, the delta-predictability disagreement across the two draws is compatible with noise, but noise was not separately identified or tested from only two draws.

5 Limitations

This exploratory study is limited to one topic, the Qwen3 family, gradient-based methods, and paired training seeds for the declared families. Two-seed replication applies only to declared seed-paired contrast families. Other readings are labeled separately.

The delta-predictability disagreement across seeds is compatible with noise at two draws, but noise was neither separately identified nor tested from so few draws. A small denominator offers a plausible, possible explanation for wider attributable-fraction intervals and sign changes across seeds, but this account is untested and non-causal.

Excluding the alternative that perturbing the corpus at all moves a small quantity would require an unrun, dose-matched random-removal arm retrained under the same procedure.

The surface-form comparison confounds length with premise density, so their contributions cannot be separated. A formulation that sums gradient agreement along the training trajectory could score differently; comparison with the evaluated formulation lies outside this evaluation and remains speculative.

6 Conclusion

The removal results distinguish potential removability from reliable attribution: at the wider budget, ranking sources by separately measured per-source effects was consistent with removability in the declared cells, whereas narrow-budget TracIn removal increased the installed effect across both seeds. Here, attributable fraction denotes the share of the installed pool effect removed after retraining, so negative values indicate an increase after removal. The oracle evidence remains cell-specific and does not establish general causal removability.

For the methods and cells displayed in the overlap table, no method separated from the word-count baseline under the specified interval-overlap criterion. This conclusion is restricted to those displayed methods and cells, and interval overlap is not a pairwise test.

7 Appendix: Full contrast ledger

Across the full contrast ledger, the fixed tables remain authoritative for numerical estimates and confidence intervals.

A Supplementary diagnostics

Table 2: Complete ledger of every declared contrast, including installed effects, ranking diagnostics, pool and control readings, and attributable fractions. Point estimates, available 95% intervals, and recorded qualifications are shown without pooling unlike readings. Source: paper/synthesis.json. *Note.* Rows without intervals are deterministic point contrasts over recorded arm scores.

intervention	net effect	qualification
Installed effects		
Installed effect: descriptive conclusions (md)	+0.11 [+0.087, +0.14]	Conclusion-stating corpus at matched dose.
Installed effect: premises as ~740-word articles (mev)	+0.0072 [+0.0016, +0.014]	Two-epoch reading, matched off-topic control retrained per seed.
Installed effect: the SAME premises as ~105-word answers (ms)	+0.16 [+0.13, +0.19]	Identical premise specification to the row above; only surface form differs.
Installed effect: long articles, dense premises	+0.055 [+0.039, +0.071]	Same premise density as the row above at ~700 words: the length factor alone.
Installed effect: short answers, dense premises	+0.12 [+0.090, +0.15]	Third person, ~103 words, ~15% premise density. With the row below, isolates length at fixed density.
Installed effect: short answers, sparse premises	+0.051 [+0.031, +0.072]	Same length as the short-dense row at ~4% density: the density factor alone.
Installed effect: explicit stance (me)	+0.31	Explicit-stance ceiling; token dose is lower than the evidence corpus. Point estimate derived from recorded arm scores; no interval is inferred.
Pool and controls		
Pool effect before removal, seed 42 (dB_before)	+0.017 [+0.0084, +0.026]	Denominator of every AF at this seed; full fine-tuning.
Pool effect before removal, seed 7 (dB_before)	+0.025 [+0.014, +0.037]	Replicates seed 42 with overlapping intervals; no LoRA-style halving.
Off-topic control machinery, seed 42	-0.00042 [-0.0034, +0.0029]	Straddles zero; full-FT control is far cleaner than the LoRA control it replaces.
Attributable fraction, by method		
AF, oracle removal at 10% budget, seed 42	-0.10 [-0.95, +0.35]	Straddles zero. The point estimate is below the 20% budget's, but these wide intervals do not establish an ordering; report as a point-estimate observation only, never as a demonstrated sublinear response.
AF, oracle removal at 10% budget, seed 7	+0.30 [-0.016, +0.55]	Seed-seven companion to the sublinear-response row.
AF, oracle removal at 20% budget, seed 42	+0.52 [+0.17, +0.86]	Removal by separately measured per-source effect; the ceiling.
AF, oracle removal at 20% budget, seed 7	+0.62 [+0.29, +0.84]	Seed-seven replication of the ceiling; spread 0.10.
AF, delta-predictability at 10% budget, seed 42	-0.58 [-1.67, -0.018]	Excludes zero NEGATIVE; contradicts the same cell at seed 7. Not a working repair.
AF, delta-predictability at 10% budget, seed 7	+0.34 [+0.078, +0.55]	Excludes zero POSITIVE; contradicts seed 42. The pair is a seed contradiction, not an average.
AF, delta-predictability at 20% budget, seed 42	+0.13 [-0.80, +0.59]	Straddles zero at the wider budget.
AF, delta-predictability at 20% budget, seed 7	+0.24 [-0.30, +0.62]	Straddles zero; the wider budget does not resolve the seed contradiction at 10%.
AF, TracIn removal at 10% budget, seed 42	-0.91 [-2.15, -0.28]	Negative and excluding zero: removing top-ranked documents INCREASED the effect.
AF, TracIn removal at 10% budget, seed 7	-0.42 [-1.09, -0.022]	Seed-seven replication of the counterproductive direction.
AF, TracIn removal at 20% budget, seed 42	-0.47 [-1.51, +0.19]	Straddles zero at the wider budget.

intervention	net effect	qualification
AF, TracIn removal at 20% budget, seed 7	+0.17 [-0.38, +0.52]	Straddles zero; completes the TracIn cell block.
AF, TracIn-cosine removal at 10% budget, seed 42	-0.95 [-2.32, -0.32]	The prescribed length-normalisation fix; also counterproductive at this seed and budget.
AF, TracIn-cosine at 10% budget, seed 7	-0.20 [-0.77, +0.16]	Straddles zero at this seed; the counterproductive reading is single-seed for this method.
AF, TracIn-cosine removal at 20% budget, seed 42	-0.053 [-0.76, +0.33]	Straddles zero at the wider budget.
AF, TracIn-cosine at 20% budget, seed 7	+0.22 [-0.17, +0.48]	Straddles zero; completes the TracIn-cosine cell block.
AF, word-count baseline at 10% budget, seed 42	-0.49 [-1.53, +0.18]	The model-free baseline. Non-separation is asserted as interval OVERLAP with each content-keyed method in the same budget and seed cell; no pairwise contrast was computed and none may be implied.
AF, word-count baseline at 10% budget, seed 7	-0.14 [-0.68, +0.22]	Baseline replication; the same interval-overlap criterion applies.
AF, word-count baseline at 20% budget, seed 42	-0.30 [-0.94, +0.087]	Baseline at the wider budget; the same interval-overlap criterion applies.
AF, word-count baseline at 20% budget, seed 7	+0.12 [-0.37, +0.45]	Completes the baseline block. Every content-keyed cell is compared against these by interval overlap only.
Retrieval rank correlations (not attributable fractions)		
Rank correlation with installed effect: embedding retriever, negative polarity	+0.94 [+0.89, +0.94]	Same reading at the other polarity. A rank correlation over 6 sources; ranking fidelity is not removal efficacy.
Rank correlation with installed effect: embedding retriever, positive polarity	+0.89 [+0.89, +0.94]	Spearman over 6 sources, not an attributable fraction; no removal was run for this retriever. intfloat/e5-base-v2, chunked, score = max over chunks; the mean-over-chunks variant gives +0.9429 at both polarities. Not length-confounded: rho(score, words) is -0.18 to -0.29.
Rank correlation with installed effect: length baseline, negative polarity	+0.26 [+0.14, +0.66]	Baseline at the other polarity. Rank correlation only.
Rank correlation with installed effect: length baseline, positive polarity	+0.26 [+0.14, +0.43]	The model-free baseline the retriever is read against, from the same bootstrap. Rank correlation only.